

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277286

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

PATEL; Ashish D. et al.

SHELF DETECTION AND CONTROL FOR VACUUM ARC REMELTING

Abstract

A vacuum arc remelting (VAR) system for forming an ingot from an electrode includes a crucible configured to accommodate the electrode and the ingot, one or more electromagnetic energy sources arranged about the crucible, and a controller configured to provide electric current to the one or more electromagnetic energy sources and adjust the electric current through the one or more electromagnetic energy source. The adjustment of the electric current is based on an approximation of a thickness of a shelf, wherein the shelf is formed based on an accumulation of debris on an inner periphery of the crucible.

Inventors: PATEL; Ashish D. (Wyomissing, PA), HERSCOVICI; Nir (Las Vegas, NV), DEBARBADILLO; John J. (Barboursville, WV), O'CONNELL; Corey (Chesapeake, OH)

Applicant: Titanium Metals Corporation (Henderson, NV)

Family ID: 95065516

Assignee: Titanium Metals Corporation (Henderson, NV)

Appl. No.: 19/068227

Filed: March 03, 2025

Related U.S. Application Data

us-provisional-application US 63560524 20240301

Publication Classification

Int. Cl.: C22B9/20 (20060101); C22B9/04 (20060101); G01B11/06 (20060101); G06T7/13 (20170101); G06T7/60 (20170101); G06V10/25 (20220101); G06V10/44 (20220101);

U.S. Cl.:

CPC **C22B9/20** (20130101); **C22B9/04** (20130101); **G01B11/06** (20130101); **G06T7/13** (20170101); **G06T7/60** (20130101); **G06V10/25** (20220101); **G06V10/44** (20220101); **G06V10/82** (20220101); G06T2207/20081 (20130101); G06T2207/20084 (20130101); G06T2207/30136 (20130101); G06V2201/06 (20220101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/560,524, filed on Mar. 1, 2024. The disclosure of the above application is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to furnaces, and more specifically to anomaly reduction for vacuum arc furnaces.

BACKGROUND

[0003] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0004] A vacuum arc remelting (VAR) process is generally used in the processing of high-performance titanium, zirconium, nickel-based alloys and steel, among other alloys. Generally, a VAR system gradually melts an electrode by an electric current that flows through the electrode and arcs to molten metal contained within a crucible. The applied melting current is varied during the process, to achieve the desired molten metal pool geometry. At times, the electric arc can cause beads of metal to spatter onto portions of the crucible wall that are above the molten metal. These portions of the crucible are cold and can solidify the beads into a porous nonhomogeneous mass, which can cause irregularities in the ingot.

[0005] These issues related to VAR ingot surface quality and internal quality, among other issues related to VAR processes, are addressed by the present disclosure.

SUMMARY

[0006] This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

[0007] An image is used to provide an indication of the quantity of spatter that has accumulated on the crucible wall. As the quantity of beads increases, as shown in the image, an electromagnetic energy source is implemented to reduce additional accumulation by adjusting the arc region used to melt the beads. For example, the magnetic fields emitted from the electromagnetic energy source may be increased based on the accumulated spatter to redirect the arc towards the surface of the ingot.

[0008] The spatter may be quantified using edge detection. Edges may be detected that indicate the spatter, the crucible, and the electrode within a portion of an image. The distances between the edges may be used to quantify the accumulation of beads on the crucible wall.

[0009] In one form of the present disclosure, a method for forming an ingot from an electrode with a system, the system including a crucible and one or more electromagnetic energy source, the method includes: melting a first portion of the electrode to form a molten pool within the crucible, wherein debris from the first portion of the electrode accumulate on an inner periphery of the crucible and the accumulation of the debris forms a shelf; approximating a thickness of the shelf; and operating the system, wherein the operation of the system is based on the approximation of the thickness of the shelf to control the shelf thickness during the melting.

[0010] In variations of this method, which may be implemented individually or in combination: the operation of the system includes melting the electrode based on the approximation of the thickness of the shelf; the operation of the system includes adjusting a gap between the electrode and the molten pool, wherein the adjustment of the gap is based on the thickness of the shelf; the gap is based on a weight of the ingot and a vertical position of the electrode; the operation of the system includes adjusting electric current through the one or more electromagnetic energy source, the adjustment of the electric current is based on the approximation of the thickness; the adjustment of electric current increases a root sum of squares value of the electric current over time; the approximation of the thickness is based on steps including: selecting a region of interest, wherein the region of interest is based on an image and the region of interest includes a depiction of an annular region between the electrode and the crucible; identifying first data based on the region of interest, wherein the first data is indicative of a first edge; identifying second data based on the region of interest, wherein the second data is indicative of a second edge; and determining a distance based on the first data and the second data; the distance is indicative of the thickness of the shelf as a percentage of the annular region between the electrode and the crucible occupied by the shelf; the region of interest includes pixel data and wherein the distance is based on a quantity of one or more pixels between a first pixel of the first data and a second pixel of the second data; the distance is based on a column of pixels or a row of pixels; the distance is a radial distance; the region of interest includes pixel data and wherein the identification of the first data is based on steps including removing hue information from the pixel data; the region of interest includes pixel data and wherein the identification of the first data is based on steps including applying a filter to the pixel data; the application of the filter removes noise between pixels of the pixel data, the application of the filter determines a gradient associated with the pixel data, or the application of the filter converts a pixel of the pixel data to a maximum value or a minimum value; the application of the filter is part of a thresholding algorithm; the application of the filter is part of an edge detection algorithm; the first edge is defined by an outer periphery of the electrode or the inner periphery of the crucible; the second edge is defined by one or more of the debris; the first data includes a first pixel within the region of interest and the second data includes a second pixel within the region of interest, and wherein the distance is based on a quantity of pixels between the first pixel and the second pixel; one or more portion of the first edge is continuous or one or more portion of the second edge is continuous; one or more portion of the first edge is discontinuous or one or more portion of the second edge is discontinuous; the selection of the region of interest is based on a neural network, and the neural network includes weights trained to recognize the region of interest based on a corpus of training images; the corpus of training images includes depictions of annular regions similar to the annular region; the depiction of the annular region between the electrode and the crucible includes a maximum distance between the electrode and the crucible within the image; and the system includes a camera and the method further includes: capturing the image with the camera; and sending the image to a network video recorder.

[0011] In another form of the present disclosure, a method for training a neural network to select a region of interest, wherein the region of interest includes a depiction of an annular region between an electrode and a crucible, the method includes: determining a corpus of training images, wherein the training images include predetermined bounding boxes and wherein the region of interest is similar to portions of the images encompassed by the predetermined bounding boxes; revising weights of the neural network, wherein the revision of the weights is based on a first subset of the corpus of training images and the revision of the weights is expected to reduce an error between the selection of the region of interest and the predetermined bounding boxes; and validating the neural network, wherein the validation of the neural network is based on a second subset of the corpus of training images.

[0012] In variations of this method, which may be implemented individually or in combination: the neural network includes one or more convolutional layer; the revision of the weights of the neural

network is repeated until the validation of the neural network exceeds a predetermined value of loss; the neural network outputs a tensor including a location and a size of the region of interest; and the region of interest includes pixel data.

[0013] In another form of the present disclosure, a vacuum arc remelting (VAR) system for forming an ingot from an electrode, the system includes: a crucible configured to accommodate the electrode and the ingot; one or more electromagnetic energy sources arranged about the crucible; and a controller configured to: provide electric current to the one or more electromagnetic energy sources, and adjust the electric current through the one or more electromagnetic energy source, wherein the adjustment of the electric current is based on an approximation of a thickness of a shelf, wherein the shelf is formed based on an accumulation of debris on an inner periphery of the crucible.

[0014] In variations of this system, which may be implemented individually or in combination: the controller includes: one or more processor; and one or more non-transitory computer-readable medium, wherein the one or more non-transitory computer-readable medium including instructions executable by the one or more processor cause the controller to: provide the electric current, and adjust the electric current through the one or more electromagnetic energy source; the one or more electromagnetic energy source is toroidal, and the system further includes an axial electromagnetic energy source wound about a longitudinal axis of the crucible; the one or more electromagnetic energy source includes a first electromagnetic energy source and a second electromagnetic energy source; and the one or more electromagnetic energy source and the crucible are configured to move relative to one another along a longitudinal axis of the crucible.

[0015] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] In order that the disclosure may be well understood, there will now be described various forms thereof, given by way of example, reference being made to the accompanying drawings, in which:

[0017] FIG. 1 is a schematic view of a vacuum arc remelting furnace in accordance with the teachings of the present disclosure;

[0018] FIG. 2A is a perspective view of a support assembly constructed in accordance with the teachings of the present disclosure;

[0019] FIG. 2B is a side view of the support assembly of FIG. 2A, illustrating a crucible mounted therein;

[0020] FIG. 3A is a partial top view of a vacuum arc remelting furnace in accordance with the teachings of the present disclosure;

[0021] FIG. 3B is a partial top view of an electromagnetic energy source of a vacuum arc remelting furnace in accordance with the teachings of the present disclosure;

[0022] FIG. 4A is a schematic of a controller architecture in accordance with the teachings of the present disclosure;

[0023] FIG. 4B is a schematic of a primary controller in accordance with the teachings of the present disclosure;

[0024] FIG. 4C is a schematic of a secondary controller in accordance with the teachings of the present disclosure;

[0025] FIG. 5 is a graph illustrating the current in each of the primary electromagnetic energy source and the secondary electromagnetic energy source being in phase in accordance with the

teachings of the present disclosure;
[0026] FIG. 6A is a side view of a crucible and debris in accordance with the teaching of the present disclosure;
[0027] FIG. 6B is a top view of the crucible and one or more camera in accordance with the teachings of the present disclosure;
[0028] FIG. 7A is a side view of an ingot and an accumulation of debris in accordance with the teachings of the present disclosure;
[0029] FIG. 7B is a perspective view of an ingot and an accumulation of debris in accordance with the teachings of the present disclosure;
[0030] FIG. 8A depicts an image from one camera in accordance with the teachings of the present disclosure;
[0031] FIG. 8B depicts an image from one camera in accordance with the teachings of the present disclosure;
[0032] FIG. 8C depicts an image from one camera in accordance with the teachings of the present disclosure;
[0033] FIG. 8D depicts an image from one camera in accordance with the teachings of the present disclosure;
[0034] FIG. 9A depicts a region of interest in accordance with the teachings of the present disclosure;
[0035] FIG. 9B depicts a region of interest in accordance with the teachings of the present disclosure;
[0036] FIG. 9C depicts a region of interest in accordance with the teachings of the present disclosure;
[0037] FIG. 9D depicts a region of interest in accordance with the teachings of the present disclosure;
[0038] FIG. 10 depicts pixel data and individual pixels of a region of interest in accordance with the teachings of the present disclosure;
[0039] FIG. 11 depicts a portion of a region of interest in accordance with the teachings of the present disclosure;
[0040] FIG. 12A depicts a filter in accordance with the teachings of the present disclosure;
[0041] FIG. 12B depicts a filter in accordance with the teachings of the present disclosure;
[0042] FIG. 12C depicts a filter in accordance with the teachings of the present disclosure;
[0043] FIG. 13 is a diagram of a system in accordance with the teachings of the present disclosure;
[0044] FIG. 14 is a neural network in accordance with the teachings of the present disclosure;
[0045] FIG. 15 is a method in accordance with the teachings of the present disclosure; and
[0046] FIG. 16 is a subprocess in accordance with the teachings of the present disclosure.
[0047] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

DETAILED DESCRIPTION

[0048] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

[0049] Referring to FIG. 1 a vacuum arc remelting (VAR) system **100** (also commonly referred to as a VAR furnace) is shown in accordance with one or more implementations of the present disclosure. The VAR system **100** is used to gradually melt an electrode **102** to form a pool of molten metal (e.g., molten pool **103**) during the remelting process, which cools to form an ingot **104**, described in greater detail below. The VAR system **100** includes a crucible assembly **106** that is configured to accommodate the electrode **102** and the ingot **104**. As further shown, the crucible assembly includes an upper end portion **162** and a lower end portion **164**.

[0050] The crucible assembly **106** accommodates the electrode **102** and holds the ingot **104** formed

from the electrode **102**. The crucible assembly **106** includes a crucible **112** and a cooling member **114** that defines a chamber **116** around the crucible **112** for receiving a coolant, such as water, to reduce the temperature of the crucible **112**. Other suitable systems for cooling the crucible may also be used and are within the scope of the present disclosure.

[0051] During the remelting process, electrical arcs that function to melt the electrode **102** extend between the electrode **102** and the molten pool **103**, or the ingot **104**, generally defining an arc region **105** as shown. As described herein, the magnetic fields generated by an electromagnetic energy source are localized to this arc region **105** or the magnetic fields generated by an electromagnetic energy source are generated near the arc region **105**, pushing the arc outwards towards the crucible **112** thereby resulting in improved surface quality of the ingot **104** and reducing the weight of electrode remnant at the end of the remelting process. Additional details regarding systems for localizing the electromagnetic energy source are illustrated and described in U.S. Pat. No. 11,434,544, which is commonly owned with the present application and the contents of which are incorporated herein by reference in their entirety.

[0052] For example, an electromagnetic energy source (e.g., electromagnetic energy source **120**) is translated along a longitudinal axis **166** at a height/location associated with the arc region. The height may be based on a location of the ingot, which may be determined based on a weight of the ingot. In addition, a conventional axial winding **170** with respect to longitudinal axis **166** is disposed between the electromagnetic energy source **120** and the crucible assembly **106**. This conventional axial winding **170** creates an electromagnetic field along the longitudinal axis **166** for stirring the molten pool **103** and for constricting the arc below the electrode **102**. Or moving the arc away from the crucible.

[0053] The height of the electromagnetic energy source (e.g., electromagnetic energy source **120**) with respect to a centroid of the electromagnetic energy source and the bottom of the crucible **112** may be the same or nearly the same as the height of the arcing surface of the electrode **102** (e.g., the bottom surface of the electrode **102**) with respect to the bottom of the crucible **112**. As the electrode **102** melts to form the ingot **104**, the electromagnetic energy source **120** is translated along axis **166** to maintain a similar height. It should be appreciated that the speed of translation may be faster or slower than the change in height of the bottom of the electrode **102** with respect to the crucible **112**. The electromagnetic energy source **120** may also be aligned with the gap between the electrode **102** and the ingot **104**, the molten pool **103**, the ingot **104**, or somewhere there between and translate along the axis **166** as the electrode **102** melts. The electromagnetic energy source **120** is also aligned above the bottom melting surface of the electrode **102**. The electromagnetic energy source **120** is shown situated on a carriage platform **150**. The carriage platform **150** includes a sensor **152** (e.g., a distance sensor for measuring quantity of travel or position of the carriage platform **150** and electromagnetic energy source **120**). The platform is part of a linear drive assembly configured to raise and lower the electromagnetic energy source along the longitudinal axis **166**, which is described in greater detail below.

[0054] The electromagnetic energy source **120** may include windings **122**. For example, wires are wound around a core **124** to form an electromagnet. The wires may be insulated to prevent cross conduction between windings. Additional sets of windings may be layered among winding **122** to form distinct current paths, allowing for an increased electromagnetic field energized from power sources (e.g., controllers, drivers) of predetermined capacities (e.g., maximum current rating). In one form, the electromagnetic energy source **120** has a toroidal or substantially toroidal shape. For example, cores (e.g., core **124**) are toroidal or substantially toroidal and form one or more portions of a toroid. The windings (e.g., windings **122**) of the electromagnetic energy source **120** may also be toroidal and form one or more portions of a toroid. These and other variations of electromagnetic energy should be construed as falling within the scope of the present disclosure.

[0055] Referring to FIGS. 2A-2B, a support assembly **200** is shown in accordance with one or more implementations of the present disclosure. The support assembly **200** is constructed from a

non-magnetic material (e.g., stainless steel) so as to not interfere with the generated electromagnetic fields described herein. The support assembly **200** is generally configured to orient the electromagnetic energy source **120** about the crucible **112**. More specifically, the support assembly **200** is configured to mount and support both a primary electromagnetic energy source **120'** and a secondary electromagnetic energy source **202** about the crucible **112**. Further, both the primary and secondary electromagnetic energy sources **120'**, **202** are coaxial with the longitudinal axis **166** in this form of the present disclosure. The support assembly **200** further includes another platform (e.g., platform **204**) to support the secondary electromagnetic energy source **202**, which is fixed. Thus, the primary electromagnetic energy source **120'** and the crucible assembly **106** are configured to move relative to one another along the longitudinal axis **166**, while the secondary electromagnetic energy source **202** is stationary and fixed to the upper end portion **162** of the crucible assembly **106**.

[0056] The support assembly **200** generally defines a maximum travel height **220** and a minimum travel height **222** for the primary electromagnetic energy source **120'**. Thus, the secondary electromagnetic energy source **202** is generally situated beneath a barrier or floor **210**. For example, the top of the crucible **112** is accessible from a walking platform or floor **210**, which is generally made of concrete. The support assembly **200** is secured to the floor **210**, supported by stanchions or otherwise. In this manner, the secondary electromagnetic energy source **202** is configured to provide additional electromagnetic energy to melt the electrode **102** in an area that was previously unreachable by the translating primary electromagnetic energy source **120'**. The maximum travel height may be indicated by a sensor (e.g., sensor **152**) which may be situated and oriented to indicate a limit of the travel by physical contact with one or more portions of the support assembly **200**.

[0057] As further shown, the support assembly **200** may include one or more actuators (e.g., electric motor **206**, screw drive **208**, support channels) that forms an actuator assembly (e.g., a linear drive assembly). For example, the electric motor **206** drives one or more screw drives **208** configured to translate the primary electromagnetic energy source **120'**. The linear drive assembly is part of a more general lift mechanism/system, which is controlled by the vacuum arc remelting (VAR) system **100** as set forth in greater detail below.

[0058] Referring to FIG. 3A, a partial top view of a VAR system **100** is shown in accordance with one or more implementations of the present disclosure. The system **100** includes one or more electromagnetic energy sources (e.g., primary electromagnetic energy source **120'**). The primary electromagnetic energy source **120'** includes windings **122**, which may be coils (e.g., coiled wire) wrapped around the core **124**. The windings **122** are configured in segments (e.g., segments **310**, **312**, **320**, **322**, **330**, **332**). The segments are generally paired (e.g., coil pair A-A'; coil pair B-B'; and coil pair C-C'), where the coils of each pair are arranged on opposite sides of the core. For example, segment **310** is opposite segment **312** relative the crucible **112** to form a pair. Segment **320** is opposite segment **322** relative the crucible **112** to form another pair along with segment **330** being opposite segment **332** relative the crucible **112** to form another pair. Each segment **310**, **312**, **320**, **322**, **330**, **332** is insulated or non-conductive with other segments **310**, **312**, **320**, **322**, **330**, **332** and is energized with current having an opposite polarity from adjacent segments. Each coil pair is configured to receive a sinusoidal current, with the current in each pair being 120° out of phase. The direction of the magnetic field from each individual pair is represented by the arrows in FIGS. 3A-B. Once all three pairs are activated, the resultant magnetic field is rotational, with rotations per minute dictated by the time period of the sinusoidal current.

[0059] Referring to FIG. 3B, another partial top view of a VAR system **100** is shown in accordance with one or more implementations of the present disclosure, this time with the secondary electromagnetic energy source **202**, which has a similar arrangement (e.g., second set) of segments. For example, as shown, the secondary electromagnetic energy source **202** includes a core **340** and segments **350**, **352**, **360**, **362**, **370**, **372**. The segments are paired (e.g., coil pair A-A'; coil pair B-

B'; and coil pair C-C'), where the coils of each pair are arranged on opposite sides of the core). For example, segment **350** is opposite segment **352** relative the crucible **112** to form a pair. Segment **360** is opposite segment **362** relative the crucible **112** to form another pair along with segment **370** being opposite segment **372** relative the crucible **112** to form another pair.

[0060] For example, and as shown with regard to FIG. **4A**, in accordance with one or more implementations of the present disclosure, each of the primary electromagnetic energy source **120'** and secondary electromagnetic energy source **202** are powered by one or more controllers (e.g., primary controller **410**, secondary controller **420**) in a controller architecture **400**. For example, primary electromagnetic energy source **120'** is powered by the primary controller **410** and the secondary electromagnetic energy source **202** is powered by the secondary controller **420**. The primary controller **410** and the secondary controller **420** may include phase controllers (e.g., phase controllers **412**, **414**, **416**, **422**, **424**, **426**).

[0061] Although designated as distinct entities, the controllers (e.g., primary controller **410**, secondary controller **420**, phase controllers **412**, **414**, **416**, **422**, **424**, **426**) may be situated in one support assembly, board, card, or otherwise. All of the controllers described herein may be based on one or more processors. The controllers may be distributed. The controllers (e.g., phase controllers **412**, **414**, **416**, **422**, **424**, **426**) receive current (e.g., alternating current, direct current) from a power source **450**. For direct currents, the controller (e.g., phase controllers **412**, **414**, **416**, **422**, **424**, **426**) may include an H-bridge for creating an alternating current from the received direct current.

[0062] A current (e.g., direct current) is received from the power source **450**. For example, a first phase (e.g., phase A) is used to energize segments **310**, **312**, **350**, **352**. A second phase (e.g., phase B) is used to energize segments **320**, **322**, **360**, **362**. A third phase (e.g., phase C) is used to energize segments **330**, **332**, **370**, **372**. Further, additional windings may be used to increase the electromagnetic field generated by the primary and secondary electromagnetic energy sources **120'**, **202**. For example, the additional winding controller(s) **432** is used to incrementally increase the provided electromagnetic field with additional windings. The additional winding controllers **432** have the same form factor as controllers **412**, **414**, **416**, **422**, **424**, **426**, providing scalability for larger electromagnetic fields to be generated with the same hardware type.

[0063] For example, an additional winding controller **432** provides electric current for additional windings or segments **434**, **436** which may be double wound in segment A-A'. For example, the primary electromagnetic energy source **120'** includes a second winding on one or more segments (e.g., segments **310**, **312**, **320**, **322**, **330**, **332**). That is, the additional winding controller **432** provides current to a second set of windings that are wrapped in combination with windings for segment **310** and segment **312**. In such a way, additional current is used to generate larger magnetic fields by duplicating controller hardware using independent circuits.

[0064] The main controller **402** (e.g., a main VAR furnace controller) receives process parameters (e.g., a recipe) for remelting and provides status information to an HMI (Human Machine Interface). For example, the main controller **402** indicates (e.g., displays on the HMI) the position of the primary electromagnetic energy source **120** based on an indication of position from position sensor **442** (e.g., an encoder). The HMI may further depict the remelt location of the electrode **102**, the current provided to one or more of electromagnetic energy source **120**, **202**, segment **310**, **312**, **320**, **322**, **330**, **332**, **350**, **352**, **360**, **362**, **370**, **372**, **434**, **436**, or a combination thereof. The main controller **402** directs the remelt process and notifies the arc sweep controller **404** that arc position control in the crucible **112** is desired. The main controller **402**, the arc sweep controller **404**, or combinations thereof are considered a master controller as they provide instructions to controllers that are downstream (e.g., receiving commands or inputs from upstream controllers).

Communication between the main controller **402** and the arc sweep controller **404** may be provided through an Ethernet protocol. It should be understood, however, that other communications protocols, such as by way of example, transmission control protocol and internet protocol

(TCP/IP), user datagram protocol (UDP), or controller area network (CAN) protocol may be employed while remaining within the scope of the present disclosure.

[0065] The arc sweep controller **404** provides the position of the primary electromagnetic energy source, the current provided to the primary and secondary electromagnetic energy sources **120'**, **202**, alarms, or combinations thereof. The arc sweep controller **404**, or another controller, receives process inputs from one or more controller (e.g., main controller **402**, primary controller **410**, secondary controller **420**). The arc sweep controller **404**, or another controller, provides current settings to other controllers (e.g., the primary controller **410**, the secondary controller **420**), and those controllers control the output of the current allowed to flow between the power source **450** and the primary and secondary electromagnetic energy sources **120'**, **202**. The process inputs may include encoder position and current provided to the primary and secondary electromagnetic energy sources **120'**, **202**.

[0066] The arc sweep controller **404** receives an indication (e.g., an analog or digital input) provided by a limit switch (e.g., limit switch **440**). More than one indication corresponding to multiple limit switches is contemplated. For example, each limit switch **440** may be configured to provide an indication when energy to the motor controller **444** should cease, ensuring that travel of the primary electromagnetic energy source **120'** is within a predetermined range. The limit switches **440** may be disposed on the support assembly **200** at the intended maximum height and minimum height of the primary electromagnetic energy source **120** or the intended maximum height (e.g., maximum travel height **220**) and minimum height (e.g., minimum travel height **222**) of the carriage platform **150** for the primary electromagnetic energy source **120'**.

[0067] One or more position sensors (e.g., position sensor **442**) are used to determine a position of the primary electromagnetic energy source **120'** as it moves longitudinally along the crucible **112** (e.g., longitudinal axis **166**). The position sensors may be part of the electric motor **206**. For example, the position sensors (e.g., position sensor **442**) may measure an angular position in steps or otherwise of the output rotor of the electric motor. An additional sensor (e.g., sensor **152**) may be attached to, or proximate, the carriage platform **150**, which may be used alone or in combination with the other position sensors **442** to measure the position of the primary electromagnetic energy source **120'**. Position sensors may also be used to measure the angular position of one or more screw drives **208**. For example, the linear position of the screw drives **208** may be indicative of the movement of the primary electromagnetic energy source **120'** along the longitudinal axis **166** when considered in combination with the screw pitch or other information.

[0068] For example, the arc sweep controller **404** determines a numerical position (e.g., height) needed for the primary electromagnetic energy source **120'** to properly control the arc relative the bottom of the electrode **102**. The arc sweep controller **404** then converts the numerical position into a quantity of steps to advance the electric motor **206** or screw drive **208** based on the position of the position sensor (e.g., position sensor **442**) or until the position sensor reaches the desired destination. For example, the arc sweep controller **404** may be configured to track a position associated with the electrode **102** (e.g., bottom surface of the electrode **102**) with a position of the primary electromagnetic energy source **120'** using a conversion between the position sensor **442** and a position associated with the electrode **102**, continuously reducing a difference between the position of the primary electromagnetic energy source **120'** based on the position sensor and the position associated with electrode **102**. The position associated with the electrode **102** may be the ingot height, which may be derived from the ingot weight.

[0069] The primary and secondary electromagnetic energy sources **120'**, **202** may cause interference between their respective generated electromagnetic fields. For example, if the current or voltage associated with segment A of primary electromagnetic energy source **120'** (e.g., segment **310**) is out of phase with the current or voltage associated with segment A of secondary electromagnetic energy source **202** (e.g., segment **350**), the field associated with secondary electromagnetic energy source **202** may interfere, conflict, reduce, or otherwise impede the field

associated with the primary electromagnetic energy source **120'**.

[0070] Referring to FIG. **4B**, a controller is shown in accordance with one or more implementations of the present disclosure. It should be understood that the term “controller” is generally a depiction of the primary controller **410**, the secondary controller **420** (FIG. **4C**), or another controller indicated (e.g., additional winding controller **432** or phase controllers **412**, **414**, **416**, **422**, **424**, **426**). The primary controller **410** receives commands from the arc sweep controller **404**. For example, the arc sweep controller **404** may provide an indication through an output (e.g., a digital or analog voltage output) to an input **468** associated with microcontroller **460**. The microcontroller **460** receives that input as an indication to energize one or more of the phase controllers **412**, **414**, **416**. A similar control signal may be sent to the secondary controller **420** for phase synchronization as discussed with regard to FIG. **5** for energization of phase controllers **422**, **424**, **426**. Secondary controller **420** and phase controllers **422**, **424**, **426** have a similar construction to that shown with regard to primary controller **410** and phase controllers **412**, **414**, **416**. Phase controllers **422**, **424**, **426** may receive a power, current, or voltage input from controller **470** or counterpart controller, or be operated by a counterpart controller similar to microcontroller **460** from instructions provided by arc sweep controller **404**. Outputs from the phase controllers **422**, **424**, **426** are monitored and provided to the arc sweep controller **404** for control and operation of phase controllers **412**, **414**, **416**, **422**, **424**, **426**.

[0071] In this form, the microcontroller **460** is configured to operate the H-bridge of the phase controllers **412**, **414**, **416** based on control signal **462**. For example, the microcontroller operates switches of the H-bridge, or half bridge(s), to generate an alternating current from a direct current provided from source **450**. The switches may be pulse-width modulated (PWM) to form the sinusoidal alternating currents shown in FIG. **5**. For example, the microcontroller **460** operates the switches for the H-bridge associated with phase controller **412** with a 120° offset from the switches for the H-bridge associated with phase controller **414** and 240° offset from the switches for the H-bridge associated with phase controller **416**, providing three-phases as indicated in FIG. **5**. The phase controllers **422**, **424**, **426** are similarly operated to provide current waveforms **502**, **504**, **512**, **514**, **522**, **524**.

[0072] The magnitude of each peak generated by the H-bridge phase controllers **412**, **414**, **416** are controlled by controller **470**. The magnitude of each peak generated is controlled using controller **470** to balance peak currents of each phase and control the peak current output by the phase controllers **412**, **414**, **416**, **422**, **424**, **426**. For example, controller **470** receives an indication of desired power, current, voltage or combination thereof indication (e.g., indication **466**). The indication **466** may be provided as a numeral value, a percentage of total power, or otherwise. For example, the controller **470** may receive a voltage within the range of 0-10 Volts, and the controller **470** may scale the output power, current, or voltage from 0-100% based on the indication (e.g., indication **466**). The indication **466** is used to control the power, voltage, or current provided to respective H-bridges of phase controllers **412**, **414**, **416**, **422**, **424**, **426**. The indication **466** may be an analog voltage output from the arc sweep controller **404** to an analog input of controller **470** between 0.0-10 Volts and corresponding to and output of 0-400 Volts or 0-10 Amps of the controller **470**. The indication **466** may also be between 0.0-5.5 Volts and corresponding to and output of 0-400 Volts or 0-10 Amps of the controller **470**.

[0073] Taps **464** are provided to measure the power, voltage, current, or resistances of each of the coils or phases. For example, the arc sweep controller **404** may be configured to receive taps **464** as shown, providing an indication of the voltage, current, or power provided to segments **310**, **312**, **320**, **322**, **330**, **332**, **350**, **352**, **360**, **362**, **370**, **372**. Arc sweep controller **404** may be further configured to monitor a current loop based on segments **310**, **312**, **320**, **322**, **330**, **332**, **350**, **352**, **360**, **362**, **370**, **372**. For example, a current loop is formed between segments **310**, **312** for measuring a resistance associated with those segments before remelting commences. The measured resistance may be used to ensure similar current, or voltage, peaks are formed and provided to the

segments **310, 312, 320, 322, 330, 332** such that the resulting electromagnetic fields generated by the primary electromagnetic energy source **120** are balanced between segments.

[0074] For example, manufacturing and use may impart variations between segments **310, 312, 320, 322, 330, 332** and cause generated fields to have unequal magnitudes, which can result in reduced arc control or unbalanced arcing with respect to the longitudinal axis **166**. As such, the resistances of current loops are measured by the arc sweep controller **404** and used to adjust power, current, or voltage output from controller **470** based on indication **466**. The power, current, or voltage provided to segments **310, 312, 320, 322, 330, 332** may be based on the resistance of the current loop associated with one or more segments **310, 312, 320, 322, 330, 332**.

[0075] Referring to FIG. **4C**, a controller is shown in accordance with one or more implementations of the present disclosure. It should be understood that the term “controller” is generally a depiction of the primary controller **410**, the secondary controller **420**, or another controller indicated (e.g., additional winding controller **432** or phase controllers **412, 414, 416, 422, 424, 426**). The secondary controller **420** receives commands from the arc sweep controller **404**. For example, the arc sweep controller **404** may provide an indication through an output (e.g., a digital or analog voltage output) to an input **488** associated with microcontroller **480**. The microcontroller **480** receives that input as an indication to energize one or more of the phase controllers **422, 424, 426**.

[0076] In this form, the microcontroller **480** is configured to operate the H-bridge of the phase controllers **422, 424, 426** based on control signal **482**. For example, the microcontroller operates switches of the H-bridge to generate an alternating current from a direct current provided from source **450**. The switches may be pulse-width modulated (PWM) to form the sinusoidal alternating currents shown in FIG. **5**. For example, the microcontroller **480** operates the switches for the H-bridge, or half bridge(s), associated with phase controller **422** with a 120° offset from the switches for the H-bridge associated with phase controller **424** and 240° offset from the switches for the H-bridge associated with phase controller **426**, providing three-phases as indicated in FIG. **5**.

[0077] The magnitude of each peak generated by the H-bridge phase controllers **422, 424, 426** are controlled by controller **490**. The magnitude of each peak generated is controlled using controller **490** to balance peak currents of each phase and control the peak current output by the phase controllers **422, 424, 426**. For example, controller **490** receives an indication of desired power, current, voltage or combination thereof indication (e.g., indication **486**). The indication **486** may be provided as a numeral value, a percentage of total power, or otherwise. For example, the controller **490** may receive a voltage within the range of 0-10 Volts and the controller **490** may scale the output power, current, or voltage from 0-100% based on the indication (e.g., indication **486**). The indication is used to control the power, voltage, or current provided to respective H-bridges of phase controllers **422, 424, 426**. For example, secondary controller **420** may support voltages from zero to 400 Volts and currents from 0 to 10 Amps. The indication **486** may be an analog voltage output from the arc sweep controller **404** to an analog input of controller **490** between 0.0-10 Volts and corresponding to and output of 0-400 Volts or 0-10 Amps of the controller **490**. The indication **486** may also be between 0.0-5.5 Volts and corresponding to and output of 0-400 Volts or 0-10 Amps of the controller **490**.

[0078] Taps **484** are provided to measure the power, voltage, current, or resistances of each of the coils or phases. For example, the arc sweep controller **404** may be configured to receive taps **484** as shown, providing an indication of the voltage, current, or power provided to segments **310, 312, 320, 322, 330, 332, 350, 352, 360, 362, 370, 372**. The arc sweep controller **404** may be further configured to monitor a current loop based on segments **310, 312, 320, 322, 330, 332, 350, 352, 360, 362, 370, 372**. For example, a current loop (not shown) is formed between segments **350, 352** for measuring a resistance associated with those segments before remelting commences. The measured resistance may be used to ensure similar current, or voltage, peaks are formed and provided to the segments **350, 352, 360, 362, 370, 372** such that the resulting electromagnetic

fields generated by the primary electromagnetic energy source **120'** are balanced between segments.

[0079] For example, manufacturing and use may impart variations between segments **350, 352, 360, 362, 370, 372** and cause generated fields to have unequal magnitudes, which can result in reduced arc control or unbalanced arcing with respect to the longitudinal axis **166**. As such, the resistances of current loops are measured by the arc sweep controller **404** and used to adjust power, current, or voltage output from controller **490** based on indication **486**. The power, current, or voltage provided to segments **350, 352, 360, 362, 370, 372** may be based on the resistance of the current loop associated with one or more segments **350, 352, 360, 362, 370, 372**.

[0080] As shown in accordance with one or more implementations of the present disclosure, FIG. 5 illustrates the alignment of phases **500** for respective segments of the primary and secondary electromagnetic energy sources **120'** and **202**. For example, segments **310, 312, 350, 352** have aligned phases and magnitude as shown with current waveforms **502, 504** that are shown in phase or substantially in phase (e.g., less than 30° out of phase). Segments **320, 322, 360, 362** have aligned phases as shown with current waveforms **512, 514** that are shown in phase or substantially in phase (e.g., less than 30° out of phase). Segments **330, 332, 370, 372** have aligned phases as shown with current waveforms **522, 524** that are shown in phase or substantially in phase (e.g., less than 30+0 out of phase).

[0081] As the primary electromagnetic energy source **120'** nears the maximum travel height (e.g., nears the location of one or more limit switches **440**), the arc sweep controller **404** sends commands or otherwise causes the phase controllers (e.g., phase controllers **412, 414, 416**) to increase current supplied to respective segments (e.g., segments **310, 312, 320, 322, 330, 332**). Thus, the electrode **102** is continually remelted as the primary electromagnetic energy source **120'** reaches the maximum travel height **220**, as indicated by limit switch **440** or position sensor **442**. As the primary electromagnetic energy source **120'** reaches the maximum travel height **220**, the current applied to the primary electromagnetic energy source **120'** (e.g., segments **310, 312, 320, 322, 330, 332**) is increased. The increase may be exponential over ingot height (e.g., more exponential than linear) until the primary electromagnetic energy source **120'**, or segments thereof, reach magnetic field saturation.

[0082] Referring to FIGS. 6A-6B, a crucible **112** is shown in accordance with the teaching of the present disclosure. As the electrode **102** is melted by the arc region **105**, debris **600** splatter as beads and accumulate on the crucible **112**. The debris **600** may cool quickly or faster than the solidification of the molten pool **103**, accumulating, and forming a shelf **602** (e.g., crown, crest, ledge, protrusion, shoreline). The shelf **602**, or portion thereof, may break off or detach from the crucible **112** as the ingot **104** and molten pool **103** rise during remelting, causing imperfections or irregularities **604** in the ingot (e.g., defects, white spots). The shelf **602** may be an accumulation of debris **600** that is detectable as an edge. The shelf **602** may be a detectable accumulation of debris **600** that requires remediation, mitigation, or reduction.

[0083] The debris **600** may include magnesium, calcium, chromium, manganese, and other evaporates, which may accumulate on a wall of the crucible **112** and coat solid particles. The debris **600** may include nitrides and oxides that wash onto the shelf **602**. The shelf may overhang the molten pool **103** forming an undercut **608**. The undercut **608** may cause a weakness in the attachment of the shelf **602** to the crucible **112** leading to eventual departure and formation of imperfections and irregularities **604**. The molten pool **103** may be formed by pendant droplets **610** forming on the electrode **102** and falling into the molten pool **103**. The electrode **102** may be translated in a direction **612** along the longitudinal axis **166** to form a gap **606** relative the bottom of the electrode **102** and the molten pool **103**. The pendant droplets **610** may be irregular, and the gap **606** may be determined based on the stepper position of the motor controlling the position of the electrode **102** and the weight of the ingot **104**. The pendant droplets **610** may form a torus **614**. As described herein, the annular region **800** may be defined between the torus **614**, the electrode

102, or a combination thereof.

[0084] A camera (e.g., camera **650**, **652**, **654**) may be used to identify the shelf **602**, or edge indicative thereof, within the annulus. For example, the camera may be positioned to view an annulus between the crucible **112** and the electrode **102**. The cameras **650**, **652**, **654** may peer through a viewport (e.g., viewport **656**) through the crucible assembly to view the annulus.

[0085] Referring to FIG. 7A, a side view of an ingot **104** and debris **600** is shown in accordance with the teachings of the present disclosure. The debris **600** may accumulate to form a shelf **602**. The shelf **602**, or portion thereof, may separate from the ingot **104** during the formation of the ingot **104** and fall into the molten pool **103**.

[0086] Referring to FIG. 7B, a perspective view of an ingot **104** and debris **600** in accordance with the teachings of the present disclosure. The debris **600** may accumulate to form a shelf **602**. The shelf **602**, or portion thereof, may separate from the ingot **104** during the formation of the ingot **104** and fall into the molten pool **103**.

[0087] Referring to FIGS. 8A-8D, images **810**, **820**, **830**, **840** are shown in accordance with the teachings of the present disclosure. The images **810**, **820**, **830**, **840** may be captured by one or more camera (e.g., camera **650**, **652**, **654**). One or more of the images **810**, **820**, **830**, **840** may include data from more than one camera and the data maybe joined to form a composite image. The images **810**, **820**, **830**, **840** may be two-dimensional and contain pixels. The images **810**, **820**, **830**, **840** may be three-dimensional and contain voxels. The images **810**, **820**, **830**, **840** may include data for each one of the pixels. The data may comprise a location of the pixel based on an index (e.g., row, column) and hue, luminance, color information in one or more formats (e.g., Red-Green-Blue, grayscale, Hue-Saturation-Lightness).

[0088] The images **810**, **820**, **830**, **840** may be one or more frame of a motion picture or video. Frames from the video may be skipped. For example, the frames used for images **810**, **820**, **830**, **840** may be scaled back from 30 frames per second to a range from one frame per second to four frames per second. The images **810**, **820**, **830**, **840** depict the crucible **112**, the electrode **102**, and an annular region **800** between the electrode **102** and the crucible **112**. The depiction of the crucible **112** forms an inner periphery **802** (i.e., inner diameter/surface of the crucible **112**) within the image **810**, indicating of a boundary between the depiction of the crucible **112** and the annular region **800**. The depiction of the electrode **102** forms an outer periphery **804** (i.e., outer diameter/surface of the electrode **102**) within the image **810**, indicating a boundary between the depiction of the electrode **102** and the annular region **800**.

[0089] The images **810**, **820**, **830**, **840** may individually or collectively depict changes to the debris **600**, shelf **602**, or annular region **800** at an instant or over time. In order to approximate a thickness of the shelf **602**, a region of interest may be determined for the respective images (e.g., region of interest **812**, **822**, **832**, **842**). The region of interest **812**, **822**, **832**, **842** may be determined based on machine learning. For example, the region of interest **812**, **822**, **832**, **842** may be determined by a neural network, as shown in FIG. 14.

[0090] In FIGS. 9A-9D, (which are rotated clockwise 90 degrees relative to FIGS. 8A-8D), the regions of interest **812**, **822**, **832**, **842** are shown in accordance with the teachings of the present disclosure. The regions of interest **812**, **822**, **832**, **842** may be further processed to determine edges (e.g., edges **902**, **904**, **906**) within the region of interest **812**, **822**, **832**, **842**. For example, an edge detection algorithm may be used to determine edge **902** based on the inner periphery **802** of the depiction of the crucible **112** and edge **906** based on the outer periphery **804** of the depiction of the electrode **102**.

[0091] Edge **904** may be associated with the depiction of the shelf **602** within the annular region **800**. For example, edge **904** within the annular region **800** between the outer periphery **804** of the depiction of the electrode **102** and the inner periphery **802** of the depiction of the crucible **112**. The edges **902**, **904**, **906** may be continuous (e.g., adjacent pixels that form the edge from start to finish) or discontinuous. Edges **902**, **904**, **906** may be determined based on different version of the

regions of interest **812**, **822**, **832**, **842**. For example, and with respect to region of interest **822**, the edges **902**, **904**, **906** may be determined based on versions of the region of interest **822** after the application of different thresholding algorithms (e.g., Otsu) or sigma values (e.g., Gaussian filtering sigma values) discussed in further detail with regard to FIG. **16**.

[0092] Referring to FIG. **10**, individual pixels of the region of interest **822** in accordance with the teachings of the present disclosure. The pixels of the region of interest **822** are filled with respective patterns indicative of the categories associated with each of the pixels. For example, the pixels associated with the electrode **102** are patterned for designation, and pixels associated with the crucible **112** are patterned for designation. Pixels may be defined by one or more data structure (e.g., tuple, array) including location values (e.g., row, column), color information (e.g., RGB), and categorization (e.g., associated with the electrode **102**, the crucible **112**, an edge **902**, **904**, **906**, or the annular region **800**, periphery **802**, **804**, otherwise, or a combination thereof) to form data or pixel data.

[0093] Referring to FIG. **11**, a portion of the region of interest **822** is shown in accordance with the teachings of the present disclosure. The portion of the region of interest **822** depicts the thickness of the shelf **602**. For example, the edges **902**, **904**, **906** may be found using an edge detection algorithm with pixels (e.g., pixels **1102**, **1104**, **1106**, **1122**, **1124**, **1126**). Data indicative of the edges **902**, **904**, **906** may be stored in a data structure comprising pixel data that includes the location or index of the pixel (e.g., row, column) and a category of the pixel (e.g., no edge, edge **902**, edge **904**, edge **906**). For example, a pixel may be defined in a data structure (e.g., PANDAS DATAFRAME) as [row index, column index, Red, Green, Blue, type] of the image **820** or region of interest **822** (e.g., [5, 10, 255, 255, 255, edge **902**]).

[0094] Relative distances **1108**, **1110**, **1128**, **1130** may be used to determine a thickness of the shelf **602**. For example, the distance between each pixel of the edge **904** may be determined with respect to either the edge **902** associated with the crucible **112** or the edge **906** associated with the electrode **102**.

[0095] The distance **1108** is determined with respect to pixel **1102** and pixel **1104** according to the vertical column of pixels. Distance **1108** may be inclusive of pixel **1102** and pixel **1104** (e.g., pixel distance of four) or exclusive (e.g., pixel distance of two). The distance **1110** is determined with respect to pixel **1104** and pixel **1106** according to the vertical column of pixels. Distance **1110** may be inclusive of pixel **1104** and pixel **1106** (e.g., pixel distance of nine) or exclusive (e.g., pixel distance of seven).

[0096] The distance **1128** is determined with respect to pixel **1122** and pixel **1124** according to the vertical column of pixels. The distance **1128** may be inclusive of pixel **1122** and pixel **1124** (e.g., pixel distance of five) or exclusive (e.g., pixel distance of three). The distance **1130** is determined with respect to pixel **1124** and pixel **1126** according to the vertical column of pixels. The distance **1130** may be inclusive of pixel **1124** and pixel **1126** (e.g., pixel distance of eight) or exclusive (e.g., pixel distance of six).

[0097] The average (e.g., mean, median, mode) of the pixel distances **1108**, **1128** between the edge **904** and one of the other edges **902**, **906** may be used to determine the overall percentage of the annular region **800** occupied by the shelf **602**. For example, the mean distance between edge **904** and edge **902** is 4.5 pixels and the mean quantity of pixels forming the annular region (e.g., distance between edge **902** and edge **906**) is twelve pixels, indicating that the shelf **602** occupies a percentage of 37.5% of the annular region **800** with regard to this portion of the region of interest **822** at the time that image **820** was taken. Although described with respect to vertical distances, horizontal distances or radial distances (e.g., based on the Pythagorean theorem) may be used. Maximum or minimum values of the pixel distances **1108**, **1128** are contemplated as well.

[0098] Referring to FIGS. **12A-12C**, filters **1202**, **1204**, **1206** are shown in accordance with the teachings of this disclosure. Filter **1202** is an example Gaussian filter that may remove noise between pixels of pixel data. The Gaussian filter may be various dimensions (e.g., 5×5, 3×3). The

Gaussian filter may be defined by one or more sigma values indicative of a strength of the filter. Filters **1204**, **1206** are edge detection filters (e.g., Sobel operators) for detecting edges in a first direction with filter **1204** and a second direction with filter **1206**. The filters **1204**, **1206** may be used to determine a gradient with respect to an individual pixel for an orientation of the operator (e.g., horizontal for filter **1204**, vertical for filter **1206**).

[0099] Referring to FIG. **13**, another portion of system **100** is shown in accordance with the teachings of this disclosure. The system **100** further includes a mounting bracket **1302** for camera **654**. Additional mounting brackets (not shown) are used to mount the other cameras (e.g., cameras **650**, **652**) from the crucible assembly **106**. Electrical insulation **1304** is provided to isolate the camera **654** and other electronics from the arc region **105** and other currents in and around the crucible assembly **106**. A faraday cage **1306** may also be used to prevent electromagnetic fields from reaching the camera **654** and the surrounding area. The camera **654** views the annular region **800**, crucible **112**, and electrode **102** through viewport **656**. The camera **654** may include one or more outlet wires **1309** and interfaces (e.g., serial digital interface, HDMI) to transport images from the camera **654** to junction box **1310**. The outlet wires **1309** may traverse a plug, gasket, or strain relief implement **1308**. The junction box **1310** may duplicate or split the received video signal from the camera **654** or series of images to a network video recorder **1312**. The network video recorder **1312** may store the series of images (e.g., images **810**, **820**, **830**, **840**) to form a corpus of training images (e.g., corpus of training images **1420** of FIG. **14**). The corpus of training images may be curated to ensure that they depict the annular region **800**, the crucible **112**, and the electrode **102**.

[0100] The images (e.g., images **810**, **820**, **830**, **840**) may also be sent to a computer **1314**. The computer **1314** includes circuitry and one or more processor **1316** and one or more computer-readable medium **1318**. The computer **1314** may be associated with a display **1320** and configured to provide an indication of the thickness of the shelf **602** to the main controller **402** or any other controller described herein. The computer-readable medium **1318** may include instructions that when executed by the one or more processor **1316** cause the processor **1316** to carry out one or more of the steps or actions described herein. For example, the execution may cause the processor **1316** to revise weights of a neural network **1400**, validate the neural network **1400**, select a region of interest **812**, **822**, **832**, **842**, approximate a thickness of the shelf **602**, or any other step described herein.

[0101] Referring to FIG. **14**, one form of an architecture for a neural network **1400** is shown in accordance with the teachings of this disclosure. The neural network **1400** may be implemented to determine the region of interest **812**, **822**, **832**, **842**, and the neural network **1400** may be trained on a corpus of training images **1420**. The neural network **1400** includes a series layers (e.g., **1402**, **1404**, **1406**, **1408**, **1410**). Each layer may include a combination of convolutional, pooling, and connected layers that are applied to images (e.g., image **810**, **820**, **830**, **840**) containing pixel data. The architecture for neural network **1400** may be based on a YOU ONLY LOOK ONCE architecture (e.g., YOLOv1, YOLOv5, YOLOR). A convolutional layer is a filter to detect features within an image by moving a kernel with respective weights (e.g., numerical values) in a matrix or tensor, across the resultant image for each layer (e.g., image **810**, **820**, **830**, **840** for the first layer). The weights are multiplied with each input (e.g., pixels) to create a feature map. The convolutional layers become progressively smaller along the neural network **1400** (e.g., $1280 \times 1280 \times 3$, $640 \times 640 \times 64$, $320 \times 320 \times 128$) to a fully connected layer (e.g., layer **1410**). Each convolutional layer may have respective kernel sizes (e.g., 7×7), stride (e.g., two), padding, and other parameters.

[0102] The convolutional layer may be used to detect features (e.g., the size and shape of inner periphery **802**, the size and shape of the outer periphery **804**, the size and shape of the annular region **800**) based on an image. The combination of these features over the course of the neural network **1400** results in the identification of the region of interest (e.g., region of interest **812**, **822**, **832**, **842**). In one form, an activation function or layer, such as a Rectified Linear Unit (ReLU), leaky ReLU, sigmoid, sigmoid linear unit (SiLU), or tanh, may follow the convolutional layer to

increase the nonlinearity in the output of the convolutional layer. A pooling layer is often applied after a convolutional layer or activation layer to down-sample the feature map and summarize the features. Connected layers then apply a linear transformation to the input by applying the dot product between the weights and the input. The application of these layers (e.g., layers **1402**, **1404**, **1406**, **1408**, **1410**) to produce a tensor **1412** corresponding to features of the region of interest **812**, **822**, **832**, **842**.

[0103] The neural network **1400** includes a first layer **1402**. The first layer **1402** includes a convolutional layer that is applied to the image (e.g., image **810**, **820**, **830**, **840**). The convolutional layer of the first layer **1402** may include a 7×7 kernel that moves across the input (e.g., image **810**, **820**, **830**, **840**) and is multiplied with the data of the image to form a convolved output for input into second layer **1404** or another layer. Other kernel sizes may be used as well such as a 1×1 , 2×2 , 3×3 , and so on. In one variation, an activation layer may be applied to the convolved feature map to increase the nonlinearity in the output of the convolutional layer. The first layer **1402** may also include a pooling layer to down-sample the feature map. In one form, a 2×2 max pooling layer is used, however other types (e.g., average pooling) or sizes (e.g., 3×3) may be used as well.

[0104] The neural network **1400** includes a second layer **1404** that includes another convolutional layer and pooling layer that would be applied to the output of the first layer **1402**. In one form, the convolutional layer may include a 3×3 kernel, or other appropriate size. In one variation the convolutional layer may be followed by an activation layer. In one form, the pooling layer that follows the convolutional layer or activation layer may be a 2×2 max pooling layer, however other types or sizes of pooling layers may be used as well.

[0105] Following the second layer **1404**, the neural network **1400** includes a third layer **1406**. The third layer **1406** may include multiple convolutional layers with varying kernel sizes. In one form, the third layer **1406** may include a convolutional layer with a 1×1 kernel that is applied to the output of the second layer **1404**. Following the 1×1 convolutional layer, the third layer **1406** may include another convolutional layer with a 3×3 kernel that is applied to the output of the 1×1 convolutional layer. In one form, the convolutional layers may be a series of alternating 1×1 and 3×3 convolutional layers applied successively. In one variation, the third layer **1406** may include additional convolutional layers of varying sizes and dimensions. In one variation, an activation layer, such as a ReLU or SiLU layer, may be applied after each 3×3 , 1×1 , or all convolutional layers to decrease the nonlinearity in the output of the convolutional layers. Other kernel sizes may be used as well such as a 2×2 , 4×4 , 5×5 , and so on. The third layer **1406** also includes a pooling layer following the last convolutional layer.

[0106] The neural network **1400** includes a fourth layer **1408** that is applied to the output of the third layer **1406**. The fourth layer **1408** may include multiple convolutional layers with varying kernel sizes. In one form, the fourth layer **1408** includes two convolutional layers with 3×3 kernels. In other forms, other kernel sizes may be used as well. In one variation, an activation layer may be applied after each convolutional layer. In some variations, a pooling layer may be applied after the convolutional layer.

[0107] After the fourth layer **1408**, the neural network **1400** includes a fully-connected layer **1410**. The fully-connected layer **1410** applies a linear transformation to the output of the fourth layer **1408** by applying the dot product between a tensor of weights and the output of the fourth layer **1408**. The application of the layers **1402**, **1404**, **1406**, **1408**, **1410** produce a tensor **1412** with an x-coordinate and y-pixel coordinate (e.g., [300, 400]) indicative of the center of the region of interest (e.g., region of interest **812**, **822**, **832**, **842**), a width, a height, and a confidence score. In other forms, the tensor **1412** may include other dimensions corresponding to the desired geometry, such as a radius. Other architectures for determining the region of interest **812**, **822**, **832**, **842** may be used as well, such as, but not limited to, those used in YOLO, R-CNN, Fast R-CNN, Faster R-CNN, RetinaNet, CenterNet, or another neural network trained for object detection using bounding boxes. Other architectures may be used to determine the region of interest **812**, **822**, **832**, **842** based

on the input images **810, 820, 830, 840**.

[0108] Referring to FIG. **15**, a method **1500** is shown in accordance with one or more implementations of the present disclosure. Any of the steps depicted in method **1500** may be omitted, rearranged, or duplicated.

[0109] The method **1500** may include training a neural network **1400** to select a region of interest **812, 822, 832, 842** and adjusting the melting parameters (e.g., electric current, voltage, ram position, among others) to control the melting of the electrode **102** and the thickness of the shelf **602**. The region of interest **812, 822, 832, 842** includes a depiction of the annular region **800** between the electrode **102** and the crucible **112**. In one form, the region of interest **812, 822, 832, 842** includes a depiction of the annular region **800** that has the greatest distance between the electrode **102** and the crucible **112** within the respective image **810, 820, 830, 840**. The region of interest **812, 822, 832, 842** may include a depiction of other areas of the annular region **800**, so long as the selection of the region of interest **812, 822, 832, 842** is consistent between images.

[0110] In step **1502**, a corpus of training images is determined. A corpus of training images **1420** is a collection of curated images used to train a neural network. The corpus of training images includes images similar to the images **810, 820, 830, 840** shown in FIGS. **8A-8D** with an annular region **800** between the outer periphery **804** of electrode **102** and the inner periphery **802** the crucible **112**. The images may be augmented to improve generalization by the neural network **1400**. For example, the images may be reoriented, skewed, scaled, have color adjustments, or a combination thereof. Additionally, the corpus of training images includes predetermined bounding boxes (e.g., user-defined bounding boxes establishing a ground truth) that correspond to the desired region of interest **812, 822, 832, 842**. In one form, the predetermined bounding boxes are rectangular with a width, height (e.g., 800 pixels by 1200 pixels), and center point having an x-coordinate and a y-coordinate (e.g., [300, 400]) that correspond to a predetermined coordinate system for the corpus of training images. In other variations, the bounding box may have other geometries such as, but not limited to, circular with a corresponding radius, oval, or other shapes that adequately capture the desired portion of the annular region **800**. The predetermined region of interest may always include a depiction of the annular region **800** that has the greatest distance between the electrode **102** and the ingot **104** (e.g., the widest part of the crescent).

[0111] Similar to the predetermined bounding box, the region of interest **812, 822, 832, 842** may represent a bounding box determined based on a neural network. The recognized region of interest **812, 822, 832, 842** includes a center point with an x-coordinate and a y-coordinate (e.g., [300, 400]). The region of interest **812, 822, 832, 842** may have a similar geometry as the predetermined bounding box. In one form, the region of interest **812, 822, 832, 842** can be rectangular with a width and height dimension (e.g., 800 pixels by 1200 pixels). A comparison of the resulting region of interest from the neural network **1400** may be compared with the predetermined bounding box to determine a confidence score that corresponds to an intersection over union (IOU) between the region of interest determined by the neural network and the predetermined bounding box that establishes the ground truth. This error may be used to revise the weights of the neural network **1400** based on back propagation to minimize or reduce the error in the next training evolution.

[0112] The corpus of training images **1420** is divided into a first subset (e.g., subset **1422**) and a second subset (e.g., subset **1424**). The first subset (e.g., subset **1422**) of the corpus of training images **1420** are used for revising the weights of the neural network, and the second subset (e.g., subset **1424**) of the corpus of training images **1420** are used for validating the neural network **1400**. The corpus of training images **1420** may be divided into smaller subsets to increase the number of times the neural network **1400** is trained or validated.

[0113] In step **1504**, the first subset **1422** of the corpus of training images **1420** is used to train the neural network **1400** and adjust the weights to reduce the error between the selection of the region of interest **812, 822, 832, 842** and the predetermined bounding boxes. The weights may be adjusted by using backward propagation (e.g., stochastic gradient descent) based on the error between the

predetermined bounding boxes and the predicted bounding box for the given image.

[0114] The first subset **1422** of the corpus of training images **1420**, or portion thereof, is used to train the neural network **1400**. The neural network **1400** can include a series of convolutional, pooling, and connected layers. The convolutional and connected layers have various matrices comprising weights that are applied to input data based on an image (e.g., image **810**, **820**, **830**, **840**) to produce an output. The matrices comprising weights may include a weight term to be multiplied with the input and a biasing term to be added to the resulting product. The weight term, biasing term, or both the weight and biasing term may be zero or within a range (e.g., -1 to 1). The matrices comprising weights of the convolutional layers and the connected layers may be adjusted based on the first subset **1422** of the corpus of training images **1420**. The weights are adjusted to reduce the error between the selection of the region of interest **812**, **822**, **832**, **842** and the predetermined bounding boxes.

[0115] In step **1506**, the neural network **1400** is applied to the second subset **1424** of the corpus of training images **1420** to validate the neural network **1400**. Validating the neural network **1400** with the second subset **1424** of training images evaluates the accuracy of the neural network **1400** in identifying selected region of interest **812**, **822**, **832**, **842** in relation to the predetermined bounding boxes. The revision of the weights of the neural network **1400** in step **1504** is repeated until the validation of the neural network exceeds a predetermined value of loss. In one form, the loss is determined using sum-squared error. That is the loss is equal to the summation of the squared difference between the region of interest (e.g., region of interest **812**, **822**, **832**, **842**) and predetermined bounding boxes x-coordinates, y-coordinates, widths, and heights. Other loss functions such as, but not limited to, intersection over union loss, Euclidean distance, or mean squared error can be used to evaluate the neural network.

[0116] In step **1508**, a region of interest **812**, **822**, **832**, **842** is selected. The region of interest may be based on the trained and validated neural network **1400**. Images (e.g., image **810**, **820**, **830**, **840**) including the crucible **112**, electrode **102**, and annular region **800** are captured by one or more cameras (e.g., camera **650**, **652**, **654**). The neural network is applied to the images to determine the region of interest **812**, **822**, **832**, **842**. If more than one region of interest is selected, the selected region of interest **812**, **822**, **832**, **842** will be the one with the highest confidence score. In one form, the region of interest **812**, **822**, **832**, **842** includes a depiction of the annular region **800** that has the greatest distance between the electrode **102** and the ingot **104**. The region of interest **812**, **822**, **832**, **84** may include a depiction of other areas of the annular region **800**, so long as the selection of the region of interest **812**, **822**, **832**, **84** is consistent between images.

[0117] In subprocess **1510**, a thickness of the shelf **602** is approximated. The thickness may be represented as a percentage of the annular region **800** or another geometry, quantity of pixels, an area (e.g., pixels squared, meters squared), or a physical distance (e.g., inches, millimeters). The thickness may be approximated with one or more of the steps described with regard to FIG. **16**.

[0118] In step **1512**, a system (e.g., VAR system **100**) is operated. The system is operated based on the approximated thickness of the shelf **602**. For example, an operating parameter (e.g., current, voltage, pressure, coolant flow, coolant temperature, field strength, electrode position, weight approximation, gap height) may be adjusted based on the approximated thickness of the shelf **602**. The operation of the system **100** may include melting another portion of the electrode **102**. For example, the approximation of the thickness of the shelf **602** may be used as an independent variable in the calculation or determination of another operating parameter. The approximation of the thickness of the shelf **602** may be used to augment a current voltage, pressure coolant flow, coolant temperature, field strength, electrode position, weight approximation, gap height, or another operating parameter to reduce the shelf thickness while the electrode **102** continues to be melted.

[0119] For example, the gap **606** between the electrode **102** and the molten pool **103** may be reduced or increased to reduce the shelf thickness. The gap **606** may be an average distance

between the electrode **102** and the molten pool **103** as the molten pool **103** may have a slight parabolic or paraboloidal form when viewed perpendicular to a field of view of the camera **654**. The gap **606** may be reduced by a step change, ramp change, or otherwise based on the approximation of the thickness of the shelf **602**. For example, a typical gap **606** may be five millimeters, which may be reduced or increased in proportion to the approximation of the thickness of the shelf **602**. The gap **606** may be step changed based on a predetermined quantity (e.g., one millimeter change in the gap for a approximated shelf thickness greater than a threshold).

[0120] The gap **606** may be approximated. For example, the actual gap **606** may be unknown and approximated using a vertical position of the electrode **102** (e.g., using a stepper or encoder), a weight of the ingot **104**, or a combination thereof. The weight of the ingot **104** may include a weight associated with the molten pool **103**. The gap **606** may be adjusted by a motor (e.g., stepper motor) configured to translate the electrode **102** through the crucible **112**. For example, the motor may be attached to a linear actuator to adjust a height of the electrode **102** with respect to the ingot **104**, molten pool **103**, or crucible **112** at a particular rate (e.g., millimeters/minute, inches/minute). The rate of translation of the electrode **102** may be adjusted to adjust the gap **606** over time.

[0121] The adjustments described herein may be used during the remelting of various materials (e.g., nickel alloys, titanium alloys, and steels). The gap **606** may be provisioned specifically for a type of alloy. For example, the gap **606** may be one set point for a nickel alloy and another set point for a titanium alloy. Further, furnaces may have different construction or factors that require further adjustment of the gap **606** or other operating parameters. As such, an adjustment factor may be used to offset or otherwise adjust any of the operating parameters described herein to compensate for different alloys or furnaces. Further, the shelf thickness may be converted from a percentage to an actual distance to compensate for irregularities in the crucible **112** or electrode **102** (e.g., varying radii along a length of the electrode **102**, ellipsoid deformations). The shelf **602** may have a thickness that is a linear dimension (e.g., distance from crucible) that is acted upon when the linear dimension is greater than a predetermined amount.

[0122] Step **1512** includes an adjustment to the electric current through one or more electromagnetic energy source **120'**, **202**. The electric current may be adjusted based on the thickness of the shelf **602**. For example, the electric current may be reduced or increased. The electric current may be one or more of current waveform **502**, **504**, **512**, **514**, **522**, **524**. The electric current may be through one or more of the segments **310**, **312**, **320**, **322**, **330**, **332**, **350**, **352**, **360**, **362**, **370**, **372**, **434**, **436**. For example, the electric current produces a magnetic field that controls the motion of the arc in arc region **105** based on the thickness of the shelf **602**. For example, as the thickness of the shelf **602** increases, the electric current may be increased to control the arc region **105**, reducing the debris or accumulation thereof, on the crucible **112**. In step **1514**, the electrode **102** may be further melted and an image (e.g., image **810**, **820**, **830**, **840**) from one or more camera **650**, **652**, **654** may be used to estimate the thickness of the shelf **602** resulting from the additional melting of the electrode **102**.

[0123] The electric current provided to the one or more electromagnetic energy sources **120'**, **202** may be adjusted to maintain a ratio between the percentage of the annular region **800** occupied by the shelf **602** and a measure indicative of the current through the one or more electromagnetic energy sources **120'**, **202**. For example, a current amplitude-to-percentage ratio (e.g., current amplitude of 10 to 10% occupancy) may be used. If the shelf occupies 37.5% of the annular region **800**, a current amplitude of 37.5 may be used. As another example, a case statement may be used to adjust the current amplitude based on the occupancy (e.g., 11-20% occupancy equals a current amplitude of 20, 21-30% occupancy equals a current amplitude of 30). A root sum of squares value of the current may be used instead of peak current amplitude. The electric current may be adjusted for one or more segment (e.g., segment **310**, **312**, **320**, **322**, **330**, **332**, **350**, **352**, **360**, **362**, **370**, **372**) of the one or more electromagnetic energy sources **120'**, **202**.

[0124] Referring to FIG. **16**, a subprocess **1510** is shown in accordance with one or more

implementation of the present disclosure. The subprocess **1510** provides steps that may be used to approximate the thickness of the shelf **602**. Any of the steps of subprocess **1510** may be omitted, rearranged, or duplicated. For example, in step **1602**, the region of interest (e.g., region of interest **812, 822, 832, 842**) is converted from color to greyscale. The conversion to greyscale may remove color saturation (e.g., hue) from each pixel of the pixel data. In another step **1604**, denoising or enhancements may be applied to the entire region of interest to increase the expression of edges. For example, Gaussian filter **1202** may be applied to denoise or enhance the region of interest (e.g., region of interest **812, 822, 832, 842**). Strong edges (e.g., edges **902, 906**) within the region of interest (e.g., region of interest **812, 822, 832, 842**) may be determined with a thresholding algorithm that reduces the intra-class variance between pixels (e.g., removes greyscale to change the region of interest to black and white) in step **1606**. The thresholding algorithm may be OTSU's ALGORITHM.

[0125] The edges **902, 906** may be determined in step **1608**. For example, an edge detection algorithm may apply the Gaussian filter **1202** to the thresholded pixel data. Gradients over the image may be used to determine particular edges from each direction (e.g., horizontal, vertical). For example, a filter (e.g., filters **1204, 1206**) may be used to determine each pixel of the edges **902, 906**. The filters **1204, 1206** may be an edge detection operator or kernel. Spurious edges, or indications thereof, may be removed with suppression or thresholding. Additional thresholding may be used to remove spurious edges. For all gradients over a gradient threshold, the pixel may be marked as being associated with an edge. As an example, CANNY EDGE DETECTOR may be used to determine the pixels associated with edge **902** and the pixels associated with edge **906**. Metadata may be updated to indicate which pixels are associated with each edge **902, 906**.

[0126] In step **1610**, edges associated with the shelf may be determined. For example, an edge detection algorithm may apply the Gaussian filter **1202** to the pixel data. The sigma value for the Gaussian filter **1202** applied to remove noise for the determination of the edges **904** of the shelf **602** may be different than the sigma value for the Gaussian filter **1202** used to remove noise for the determination of the edges **902, 906**. For example, the sigma value for the Gaussian filter **1202** applied to remove noise for the determination of the edges **904** of the shelf **602** may be greater than the sigma value for the Gaussian filter **1202** used to remove noise for the determination of the edges **902, 906**.

[0127] Gradients over the image may be used to determine particular edges from each direction. For example, a filter (e.g., filters **1204, 1206**) may be used to determine a gradient for each pixel of the edges **902, 906** based on the surrounding pixels. The filters **1204, 1206** may be an edge detection operator or kernel (e.g., Roberts, Prewitt, Sobel). Spurious edges, or indications thereof, may be removed with suppression or thresholding. Additional thresholding may be used to remove spurious edges. For all gradients over a gradient threshold, the pixel may be marked as being associated with an edge. As an example, CANNY EDGE DETECTOR may be used to determine the pixels associated with edge. Metadata may be updated to indicate which pixels are associated with the edge **904**. The edges **902, 904, 906** may be continuous or discontinuous over their length. For example, a discontinuous edge **904** is shown in FIGS. **9B-9D**. After the edges **902, 904, 906**, additional filtering may be used to properly express the edges. For example, the edges **902, 904, 906** may be smoothed with median filtering (e.g., replacing the color information of a given pixel with the median of the surrounding pixels). It should be appreciated that median filtering may be applied before or after any of the steps of subprocess **1510**.

[0128] In step **1612**, the image **810, 820, 830, 840** or region of interest **812, 822, 832, 842** may be checked to determine whether edges can be detected. For example, the image **810, 820, 830, 840** may be checked to determine whether all of the pixels are too bright or too dark, preventing the edges **902, 904, 906** from being determined or introducing too much noise for the edges **902, 904, 906** to have value. If the brightness of the image **810** is too high or too low, the process may stop and wait for the next image (e.g., image **820**). The electric current through the one or more

electromagnetic energy source may be maintained until the next image (e.g., image **820**) is received and processed.

[0129] In step **1614**, the thickness of shelf **602** is approximated. The thickness may be represented as a percentage of the annular region **800** or another geometry, quantity of pixels, an area (e.g., pixels squared, meters squared), or a physical distance (e.g., inches, millimeters). The approximate thickness of the shelf **602** may be based on one or more distances between pixels of the region of interest (e.g., region of interest **812**, **822**, **832**, **842**). For example, the distance between each pixel of the edge **904** may be determined with respect to either the edge **902** associated with the crucible **112** or the edge **906** associated with the electrode **102**.

[0130] For a vertically-oriented region of interest where edge **902** is above or below edge **904**, as shown in FIGS. **9A-9D**, iterations over columns may be used. For a horizontally oriented region of interest where edge **902** is to the left or right of edge **904**, iterations over rows may be used.

[0131] For example, in scanning columns for a vertically-oriented region of interest, the direction of arrow **1140** of FIG. **11** is used to iterate over each column (e.g., column **1136**) by selecting the representative pixel (e.g., pixel **1104**) for edge **904** that is closest to edge **902**.

[0132] That is, the detection of edge **904** may result in multiple pixels (e.g., pixels **1104**, **1134**) being detected for a specific column (e.g., column **1136**) because of the nature of greyscale pixel values used for determining the edge **904**. For example, if a gradient threshold (e.g., gradients greater than a value) is used to determine all of the edges within the annular region **800**, multiple pixels may be identified as relating to an edge for a column.

[0133] In such a circumstance, one of the pixels (e.g., one of pixels **1104**, **1134**) may be selected as representative for the column. Pixel data for each pixel (e.g., pixel **1104**, **1134**) of edge **904** being associated with a gradient greater than a value (e.g., gradient threshold) is added to a list for each column (e.g., pixel **1104**, **1134** for the column **1136** as shown in FIG. **11**). The pixel closest to edge **902** may be selected as representative of the column (e.g., pixel **1104**), which is the pixel with the maximum height for a vertically-oriented regions of interest where the edge **902** is above edge **904**. As another example, the pixel farthest from edge **902** may be selected as representative of the column (e.g., pixel **1134**), which is the pixel with the minimum height for a vertically-oriented region of interest where the edge **902** is above edge **904**. The process is repeated until a pixel distance is determined for each column across the region of interest **822** (e.g., distance **1108** for pixels **1102**, **1104**, distance **1128** for pixels **1122**, **1124**).

[0134] The average (e.g., mean, median) of the pixel distances between the edge **904** and one of the other edges **902**, **906** may be used to determine the overall percentage of the annular region **800** occupied by the shelf **602**. For example, the mean distance between edge **904** and edge **902** is 4.5 pixels and the mean quantity of pixels forming the annular region (e.g., distance between edge **902** and edge **906**) is twelve pixels, indicating that the shelf **602** occupies a percentage of 37.5% of the annular region **800** with regard to this portion of the region of interest **822** at the time that image **820** was taken. Although described with respect to vertical distances, horizontal distances or radial distances (e.g., based on the Pythagorean theorem) may be used.

[0135] Unless otherwise expressly indicated herein, all numerical values indicating mechanical/thermal properties, compositional percentages, dimensions and/or tolerances, or other characteristics are to be understood as modified by the word “about” or “approximately” in describing the scope of the present disclosure. This modification is desired for various reasons including industrial practice, material, manufacturing, and assembly tolerances, and testing capability.

[0136] As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

[0137] In this application, the term “controller” and/or “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed

analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components (e.g., op amp circuit integrator as part of the heat flux data module) that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

[0138] The term memory is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

[0139] The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general-purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled engineer or programmer.

[0140] The description of the disclosure is merely exemplary in nature and, thus, variations that do not depart from the substance of the disclosure are intended to be within the scope of the disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure.

Claims

1. A method for forming an ingot from an electrode with a system, the system comprising a crucible and one or more electromagnetic energy sources, the method comprising: melting a first portion of the electrode to form a molten pool within the crucible, wherein debris from the first portion of the electrode accumulates on an inner periphery of the crucible and accumulation of the debris forms a shelf; approximating a thickness of the shelf; and operating the system, wherein the operating of the system is based on the approximating the thickness of the shelf to control shelf thickness during the melting.
2. The method of claim 1, wherein the operating of the system comprises melting the electrode based on the approximation of the thickness of the shelf.
3. The method of claim 1, wherein the operating of the system comprises adjusting a gap between the electrode and the molten pool, wherein the adjustment of the gap is based on the thickness of the shelf.
4. The method of claim 3, wherein the gap is based on a weight of the ingot and a vertical position of the electrode.
5. The method of claim 1, wherein the operating of the system comprises adjusting electric current through the one or more electromagnetic energy source, wherein the adjustment of the electric current is based on the approximation of the thickness.
6. The method of claim 5, wherein the adjusting of electric current increases a root sum of squares value of the electric current over time.
7. The method of claim 1, wherein the approximation of the thickness is based on steps comprising: selecting a region of interest, wherein the region of interest is based on an image and the region of interest comprises a depiction of an annular region between the electrode and the crucible;

identifying first data based on the region of interest, wherein the first data is indicative of a first edge; identifying second data based on the region of interest, wherein the second data is indicative of a second edge; and determining a distance based on the first data and the second data.

8. The method of claim 7, wherein the distance is indicative of the thickness of the shelf as a percentage of the annular region between the electrode and the crucible occupied by the shelf.

9. The method of claim 7, wherein the region of interest comprises pixel data and wherein the distance is based on a quantity of one or more pixels between a first pixel of the first data and a second pixel of the second data.

10. The method of claim 9, wherein the distance is based on a column of pixels or a row of pixels.

11. The method of claim 9, wherein the distance is a radial distance.

12. The method of claim 7, wherein the region of interest comprises pixel data and wherein the identifying of the first data is based on steps comprising removing hue information from the pixel data.

13. The method of claim 7, wherein the region of interest comprises pixel data and wherein the identifying of the first data is based on steps comprising applying a filter to the pixel data.

14. The method of claim 7, wherein the first edge is defined by an outer periphery of the electrode or the inner periphery of the crucible.

15. The method of claim 7, wherein the second edge is defined by one or more of the debris.

16. The method of claim 7, wherein the first data comprises a first pixel within the region of interest and the second data comprises a second pixel within the region of interest, and wherein the distance is based on a quantity of pixels between the first pixel and the second pixel.

17. The method of claim 7, wherein the selecting of the region of interest is based on a neural network, and the neural network comprises weights trained to recognize the region of interest based on a corpus of training images.

18. The method of claim 17, wherein the corpus of training images comprises depictions of annular regions similar to the annular region.

19. The method of claim 7, wherein the system comprises a camera and the method further comprises: capturing the image with the camera; and sending the image to a network video recorder.

20. A vacuum arc remelting (VAR) system for forming an ingot from an electrode, the VAR system comprising: a crucible configured to accommodate the electrode and the ingot; one or more electromagnetic energy sources arranged about the crucible; and a controller configured to: provide electric current to the one or more electromagnetic energy sources, and adjust the electric current through the one or more electromagnetic energy sources, wherein the adjustment of the electric current is based on an approximation of a thickness of a shelf, wherein the shelf is formed based on an accumulation of debris on an inner periphery of the crucible.

21. The VAR system of claim 20, wherein the controller comprises: one or more processor; and one or more non-transitory computer-readable medium, wherein the one or more non-transitory computer-readable medium comprising instructions executable by the one or more processor cause the controller to: provide the electric current, and adjust the electric current through the one or more electromagnetic energy sources.

22. The VAR system of claim 20, wherein the one or more electromagnetic energy sources comprises a first electromagnetic energy source and a second electromagnetic energy source.

23. The VAR system of claim 20, wherein the one or more electromagnetic energy sources and the crucible are configured to move relative to one another along a longitudinal axis of the crucible.
